

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Lab 3: Environmental Predictors, Background, and Robustness

Today you will connect retained occurrence records to environmental rasters, define background as environments available within M, and test whether one ecological interpretation changes under two plausible assumptions about accessibility.

Learning goals

- Inspect the coordinate system, spatial grain, extent, names, and units of environmental rasters.
- Extract environmental values at occurrence locations.
- Distinguish background locations from confirmed absences.
- Compare geographic and environmental availability under two hypotheses about M.
- Classify an interpretation as stable, conditional, or not robust before fitting a model.

What you need

- Your course R project and private GitHub repository.
- The supplied files in data/ and analysis/03_environmental_space_background.R.
- R packages sf, terra, and ggplot2 installed before class if possible.

Part A – Guided workflow

Open the project and load the supplied data

1. **Open your course R Project.** Keep the same project used for Labs 1 and 2.
2. **Open the Lab 3 script.** Use analysis/03_environmental_space_background.R.
3. **Load the packages and a standardized occurrence dataset.** Use data/crotalus_triseriatus_retained_lab3.csv.

```
library(sf)
library(terra)
library(ggplot2)

set.seed(689)
occ <- read.csv("data/crotalus_triseriatus_retained_lab3.csv")
occ_sf <- st_as_sf(
  occ,
  coords = c("decimalLongitude", "decimalLatitude"),
  crs = 4326,
  remove = FALSE
)
```

Inspect the environmental raster

The supplied SpatRaster contains two WorldClim 2.1 bioclimatic variables for 1970-2000. BIO1 is annual mean temperature in degrees C. BIO12 is annual precipitation in millimeters. The layers use a coarse 10 arc-minute grid for fast classroom analysis.

```
env <- rast("data/crotalus_triseriatus_worldclim_10m.tif")

env
names(env)
crs(env)
res(env)
ext(env)
global(env, c("min", "max", "mean"), na.rm = TRUE)
```

Interpret the metadata: a raster has a variable definition, units, time period, spatial grain, coordinate system, and geographic extent.

Extract environments at occurrence locations

Extraction pulls the raster cell values beneath each retained point. It connects geographic observations to environmental space.

```
occ_env <- extract(env, vect(occ_sf))

occ_env <- cbind(
  occ[, c("record_id", "species", "decimalLongitude", "decimalLatitude")],
  occ_env[, c("bio1_c", "bio12_mm")]
)

summary(occ_env[, c("bio1_c", "bio12_mm")])
```

Define two hypotheses about M

A presence-background model compares occurrence environments with environments sampled from a calibration region. In BAM terms, that region should be related to M, the area accessible to the species over the relevant period. Changing M changes what the model treats as available.

Two scenarios

The small scenario uses a 50-km buffer and the broad uses a 250-km buffer. The distance is a modeling assumption.

```
small_buffer_km <- 50
broad_buffer_km <- 250

occ_projected <- st_transform(occ_sf, 32614)

make_m_area <- function(buffer_km) {
  area_projected <- st_union(
    st_buffer(occ_projected, dist = buffer_km * 1000)
  )
  st_transform(area_projected, 4326)
}

m_small <- make_m_area(small_buffer_km)
m_broad <- make_m_area(broad_buffer_km)
```

Restrict environmental availability to each M

`crop()` reduces processing to the polygon's rectangular extent. `mask()` then sets cells outside the polygon to NA.

```
env_small <- mask(crop(env, vect(m_small)), vect(m_small))
env_broad <- mask(crop(env, vect(m_broad)), vect(m_broad))
```

Sample background environments

Use the same number of background points in each scenario. This isolates the effect of the geographic definition of M rather than changing area and sample size at the same time.

```
n_background <- 1000

bg_small <- spatSample(
  env_small, n_background, method = "random",
  na.rm = TRUE, as.points = TRUE, values = TRUE
)

bg_broad <- spatSample(
  env_broad, n_background, method = "random",
  na.rm = TRUE, as.points = TRUE, values = TRUE
)
```

These are not pseudoabsences: the sampled locations describe available environments. The species may be absent, present but undetected, or unable to occupy a location for reasons not represented by these two predictors.

Compare assumptions in geographic space

Run the plotting block to create `output/background_M_comparison.png`. Confirm that both panels show occurrences, sampled background, and the appropriate M boundary.

```
png("output/background_M_comparison.png",
    width = 1800, height = 900, res = 180)
par(mfrow = c(1, 2))

plot(env_small[["bio1_c"]], main = "50-km M")
plot(vect(m_small), add = TRUE, border = "blue3", lwd = 2)
points(bg_small, pch = 3, cex = 0.45, col = "dodgerblue4")
points(vect(occ_sf), pch = 16, cex = 0.45)

plot(env_broad[["bio1_c"]],
      main = paste0(broad buffer km, "-km M"))
plot(vect(m_broad), add = TRUE, border = "firebrick3", lwd = 2)
points(bg_broad, pch = 3, cex = 0.45, col = "firebrick4")
points(vect(occ_sf), pch = 16, cex = 0.45)
dev.off()
```

Compare assumptions in environmental space

G-space is geographic space. E-space is the coordinate system formed by environmental variables. A broader M can add new environmental combinations even when occurrence values remain unchanged. The script converts background samples to data frames and creates a png plot within the output folder. In each panel, black points are occurrence environments and gray points are available environments under that M assumption.

```
ggplot(background_env, aes(x = bio1_c, y = bio12_mm)) +
  geom_point(color = "grey65", alpha = 0.35) +
  geom_point(data = occurrence_env, color = "black", alpha = 0.75) +
  facet_wrap(~scenario) +
  labs(
    x = "Annual mean temperature (degrees C)",
    y = "Annual precipitation (mm)"
  ) +
  theme_minimal(base_size = 12)
```

Important: a visual difference between occurrences and background is descriptive evidence. We can use those data to fit a response curve, make suitability estimates, or build occupancy probabilities.

Part B Evaluate robustness

Robustness analysis asks whether a biological interpretation survives plausible alternatives. Today you will vary one assumption - the geographic definition of M - while keeping the occurrence data, predictor layers, number of background points, and plotting procedure the same.

Complete these three tasks

1. **Choose another broad-M distance.** Use 150, 200, or 300 km. State what accessibility assumption the distance represents and one reason it may be wrong.
2. **Compare occurrence and background environments.** For either temperature or precipitation, decide whether occurrences appear higher, lower, or similar relative to background under each M assumption.
3. **Classify an interpretation.** Decide if your interpretation of the results are stable, conditional, or unstable and explain your choice with evidence from the figures and summary table.

Technical troubleshooting

A package is missing. Install it once in the Console, then rerun the script. Do not leave `install.packages()` active in the submitted script.

A file cannot be found. Open the R Project, run `getwd()` and `list.files()`, and confirm that `data/` and `analysis/` are directly inside the project root.

Raster values are NA. Check that the occurrence and raster coordinate systems are both WGS 84 before extraction.

The background sample changes. Keep `set.seed(689)` near the top of the script so the random sample is reproducible.

Academic integrity: The syllabus permits AI tools for learning and troubleshooting, but the submitted code, reasoning, and interpretation must be your own and understandable to you. Briefly acknowledge any AI use in a comment or README note.

References

- Fick, S. E., and Hijmans, R. J. (2017). WorldClim 2: New 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, 37, 4302-4315.
- Peterson, A. T., Soberon, J., Pearson, R. G., Anderson, R. P., Martinez-Meyer, E., Nakamura, M., and Araujo, M. B. (2011). *Ecological Niches and Geographic Distributions*. Princeton University Press. Chapters 4-5.
- WorldClim 2.1 historical climate data: <https://www.worldclim.org/data/worldclim21.html>

Assignment 3 submit Friday by midnight

Submit your private GitHub repository link in Canvas. The repository must contain the following items.

- **Reproducible Lab 3 script.** analysis/03_environmental_space_background.R runs from the project root (.Rproj file).
- **Geographic comparison.** output/background_M_comparison.png shows the occurrence records and background samples under both M assumptions.
- **Environmental-space comparison.** output/environmental_space_M_comparison.png shows occurrences relative to available environments under all M assumptions.
- **Robustness interpretation.** The README identifies one interpretation as stable, conditional, or not robust and supports that judgment with evidence.
- **Version history.** Commits make your progress legible and the current repository is pushed to GitHub.
- **README Lab 03 section.** Answer the following prompts.
 1. Describe the environmental layers, including units, spatial extent, and spatial grain.
 2. What does a background point represent, and why is it not an absence?
 3. Which broad-M distance did you select, and what biological assumption does it represent?
 4. How did changing M alter the sampled environmental space?
 5. Which interpretation is stable, conditional, or unstable, and what additional assumption would you test next?